

Expectation Decider

Probability & Statistics — Theory Report (Component A1)

Predicting whether a student passes a competitive mathematics exam, based on study hours, attendance, group-discussion participation and previous test scores — Dataset: 200 students.

1. Understanding the Basics

What is Probability?

Probability is a numerical measure, between 0 and 1, of how likely an event is to occur. A probability of 0 means the event is impossible, and a probability of 1 means it is certain. For example, in our dataset 88 out of 200 students passed the exam, so the probability that a randomly selected student passed is $88/200 = 0.44$.

Key Probability Terminology

- **Experiment / Trial** — an action with an uncertain outcome, e.g. randomly selecting one student.
- **Sample Space (S)** — the set of all possible outcomes, e.g. {Pass, Fail}.
- **Event** — a subset of the sample space we care about, e.g. "student studies more than 10 hrs/week".
- **Mutually Exclusive Events** — events that cannot both happen at once (e.g. Pass and Fail for the same student).
- **Independent Events** — events where one occurring does not change the probability of the other.
- **Conditional Probability $P(A|B)$** — the probability of A occurring, given that B has already occurred.

Three probability events from the dataset

- A = "A randomly selected student studies more than 10 hours per week."
- B = "A randomly selected student has attendance above 80%."
- C = "A randomly selected student passes the final exam."

2. Types of Events — Empirical vs Theoretical Probability

Empirical probability is calculated from observed data (relative frequency). **Theoretical probability** is calculated from reasoning about equally likely outcomes, independent of any single trial's results.

Empirical: $P(\text{Pass})$, observed directly from the data = $88/200 = 0.4400$.

Theoretical: assuming equally likely selection, $P(\text{study_hours} > 10) = 74/200 = 0.3700$ (classical / a-priori probability, by counting favourable outcomes over total outcomes).

3. Random Variable & Probability Distribution

Let X = "Number of students who pass the final exam, out of 3 randomly selected students (without replacement)". Since the sample is drawn without replacement from a finite population, X follows a **Hypergeometric distribution**:

$$P(X = x) = \frac{C(K, x) \cdot C(N-K, 3-x)}{C(N, 3)}, \quad x = 0, 1, 2, 3$$

where $N = 200$ (population), $K = 88$ (students who passed), $n = 3$ (sample size).

X (students who pass)	P(X)
0	0.1735
1	0.4165
2	0.3264
3	0.0836

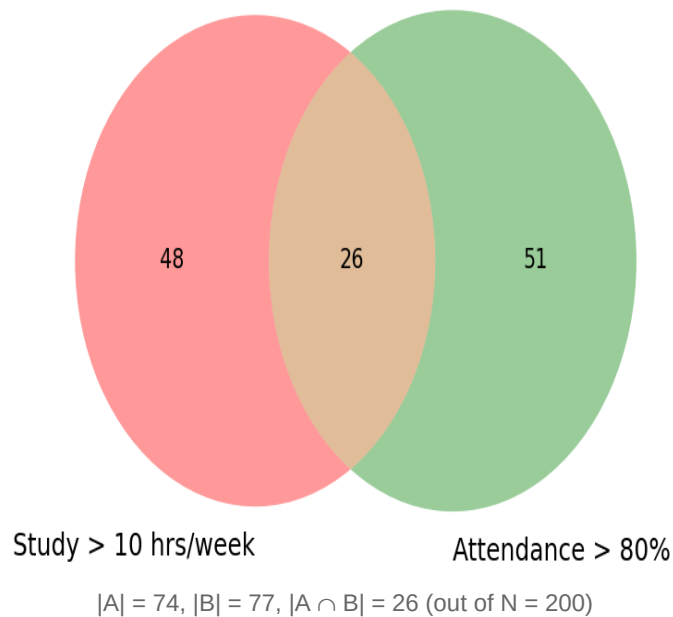
Mean $E[X] = 1.3200$ Variance $\text{Var}[X] = 0.7318$ Std. Dev = 0.8554

Interpretation: On average, about 1.32 out of every 3 randomly chosen students are expected to pass, with a variance of 0.7318 describing the typical spread of that count.

4. Venn Diagram in Probability

Set A = students who study more than 10 hrs/week. Set B = students who attend more than 80% of classes. The overlap $A \cap B$ contains students satisfying both conditions.

Venn Diagram: Study Hours vs Attendance



5. Contingency Table & Probability Calculations

Cross-tabulating group_discussion (Yes/No) against final_exam_pass (Pass/Fail):

	Fail	Pass	Total
No	50	29	79
Yes	62	59	121
Total	112	88	200

Joint probability $P(\text{Group Discussion} = \text{Yes} \cap \text{Pass}) = 0.2950$

Marginal probability $P(\text{Pass}) = 0.4400$

Conditional probability $P(\text{Pass} \mid \text{Group Discussion} = \text{Yes}) = 0.4876$

6. Understanding Relationships

Intuition: $P(\text{Pass} \mid \text{Group Discussion} = \text{Yes})$ restricts our attention only to the students who participated in group discussions, then asks what fraction of that smaller group passed. It updates our belief about a student's chances once we know they joined group discussions.

Independence check: A and B are independent iff $P(A \cap B) = P(A) \cdot P(B)$. Here $P(\text{Yes} \cap \text{Pass}) = 0.2950$, while $P(\text{Yes}) \cdot P(\text{Pass}) = 0.2662$. Since these are not equal, the events are **dependent** (and not mutually exclusive, since the joint probability is greater than 0).

7. Bayes' Theorem Application

Given (historical data):

- $P(\text{High Attendance} \mid \text{Pass}) = 0.70$
- $P(\text{High Attendance} \mid \text{Fail}) = 0.40$
- $P(\text{High Attendance}) = 0.60$
- Assume $P(\text{Pass}) = P(\text{Fail}) = 0.50$ (prior)

$$\begin{aligned} P(\text{Pass} \mid \text{High Attendance}) &= [P(\text{High Attendance} \mid \text{Pass}) \times P(\text{Pass})] / P(\text{High Attendance}) \\ &= (0.70 \times 0.50) / 0.60 = \mathbf{0.5833} \quad (58.33\%) \end{aligned}$$

Interpretation: Knowing that a student has high attendance raises the probability that they passed from a 50% prior to about 58.3% — attendance is meaningful evidence in favour of passing.

Final Summary

Across all seven tasks, the analysis of the 200-student dataset tells a consistent story:

- Study hours, attendance and previous test scores are all positively associated with passing — students above the study-hours and attendance thresholds pass at noticeably higher rates than the overall average.
- Group discussion participation is dependent on passing: participants passed at a higher rate (48.8%) than non-participants (36.7%).
- The hypergeometric distribution of passes among 3 randomly chosen students has a mean of 1.32, consistent with the overall pass rate of 44%.
- Bayes' theorem confirms attendance is informative evidence: it shifts the probability of having passed from a 50% prior up to 58.3%.

Conclusion: Among the recorded factors, attendance and study hours show the strongest positive relationship with exam success, with group discussion participation acting as an additional, statistically dependent, supporting factor.